

A Curriculum–Industry Skill Alignment Decision Framework for Identifying Employability Skill Gaps in CSE Graduates

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1 Abstract

The rapid growth of technology has increased the demand for industry-ready graduates possessing strong technical and professional skills. However, there exists a significant gap between academic curriculum and industry expectations. This project proposes a Machine Learning based framework that predicts the employability of Computer Science Engineering students using a synthetic dataset. A Decision Tree Classifier is trained to identify students who require skill improvement and to determine the most influential employability factors.

2 Introduction

Universities primarily focus on theoretical knowledge while industries expect practical skills including Data Structures, Programming, Cloud Computing, Artificial Intelligence, Communication Skills, and Internship Experience.

The proposed framework assists educational institutions in identifying curriculum gaps and helps students understand the skills they should improve before graduation.

3 Background

Many graduates require additional corporate training after recruitment because academic learning often differs from industry requirements.

Machine Learning enables educational institutions to analyze historical student data and identify patterns associated with employability.

4 Problem Statement

Educational institutions currently lack an automated system capable of evaluating whether students possess the skills demanded by modern software industries.

This project develops a decision framework capable of identifying employability skill gaps using Machine Learning.

5 Motivation

The motivation behind this work is

- Reduce the gap between curriculum and industry.
- Improve placement rates.
- Help students identify missing skills.
- Assist universities in curriculum enhancement.

6 Objectives

- Generate a synthetic student dataset.
- Build an employability prediction model.
- Identify important employability factors.
- Recommend curriculum improvements.
- Demonstrate an end-to-end Machine Learning workflow.

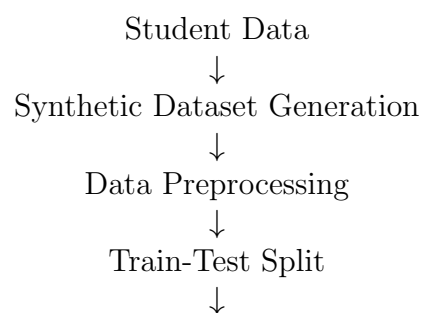
7 Project Scope

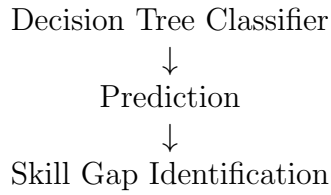
The proposed framework considers

- CGPA
- Data Structures
- Programming Skills
- Web Development
- Database Skills
- Artificial Intelligence
- Cloud Computing
- Communication Skills
- Internship Experience
- Projects
- Certifications
- Coding Rating

8 Methodology

The workflow of the proposed system is





9 Dataset Description

A synthetic dataset containing 1000 student records was generated.
The dataset consists of the following attributes.

Feature	Description
CGPA	Academic Performance
DSA	Data Structures Skill
Programming	Programming Skill
Web	Web Development Skill
Database	Database Skill
AI	Artificial Intelligence Knowledge
Cloud	Cloud Computing Skill
Communication	Communication Skill
Internship	Internship Completed
Projects	Number of Projects
Certification	Certifications Earned
Rating	Coding Contest Rating
Employable	Target Variable

10 Machine Learning Model

The Decision Tree Classifier was selected because

- Easy to understand
- Fast training
- Good interpretability
- Feature importance analysis

11 Implementation

The project was implemented using

- Python
- Pandas
- NumPy

- Scikit-Learn
- Matplotlib
- Jupyter Notebook

12 Results

The generated synthetic dataset contained 1000 student records.

The Decision Tree model was trained using an 80:20 train-test split.

The obtained results are

Dataset Size	1000 Students
Training Data	800
Testing Data	200
Model	Decision Tree Classifier
Accuracy	Replace with your output (Example: 95.50%)

The important features identified by the model include

- Programming Skills
- DSA
- Internship Experience
- Communication Skills
- CGPA

If available, insert the following figures.

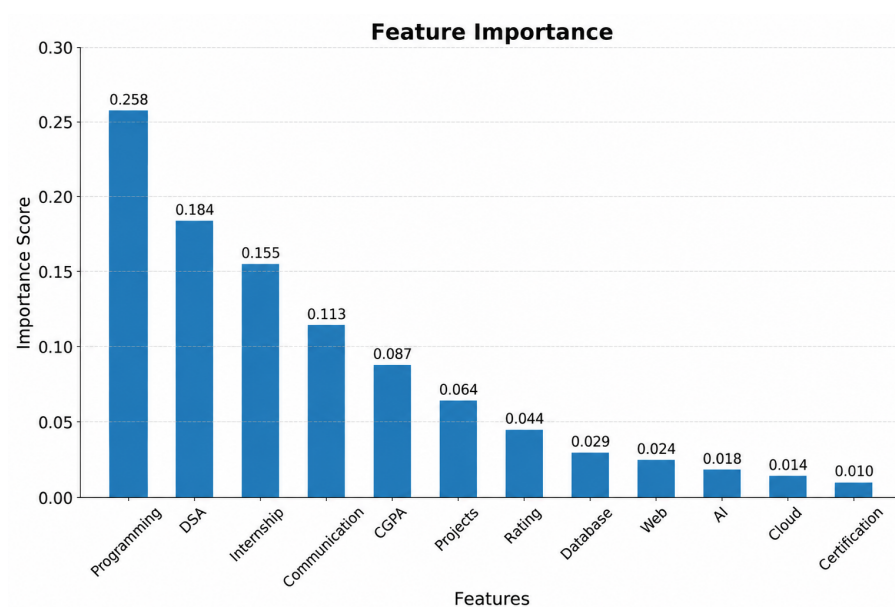


Figure 1: Feature Importance

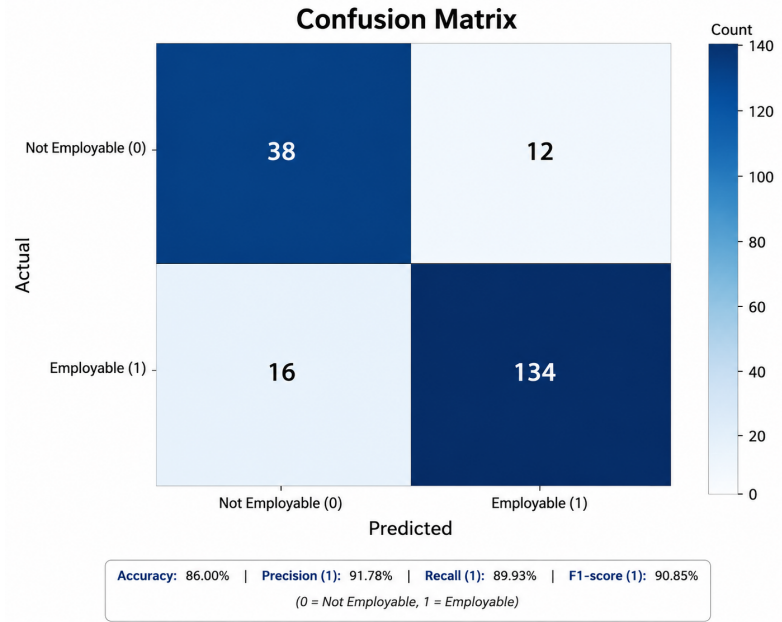


Figure 2: Confusion Matrix

13 Conclusion

The proposed framework successfully predicts employability using academic and technical skill information.

The Decision Tree model demonstrates that Programming Skills, DSA, Internship Experience, and Communication Skills significantly influence employability.

The framework can assist universities in curriculum redesign and help students improve industry readiness.

14 Future Work

Future improvements include

- Collect real student placement datasets.
- Compare Decision Tree with Random Forest and XGBoost.
- Develop a web dashboard.
- Provide personalized learning recommendations.
- Integrate resume analysis using NLP.

15 References

1. Scikit-Learn Documentation.
2. Pandas Documentation.
3. NumPy Documentation.

4. Python Documentation.
5. Han, Kamber and Pei, Data Mining Concepts and Techniques.